Active Reconnaissance

Introduction

Active reconnaissance differs from passive reconnaissance, as it is explained as a method of gaining information about a target by directly connecting to it or by making some kind of contact with the target. The module covered and discussed the use of simple tools such as; ***ping, traceroute, telnet,*** and ***nc*** used to gather information about the network, system and services.

Objectives

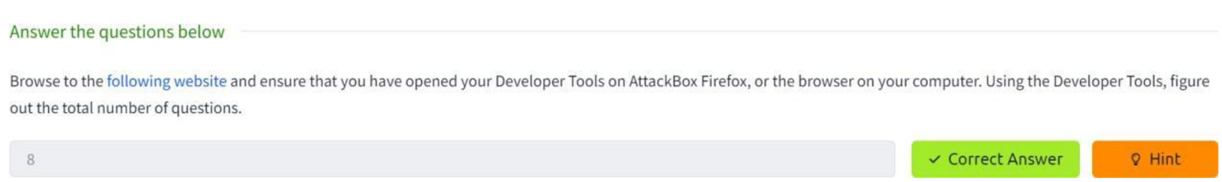
- To understand web browser as a tool
- To understand ping tool
- To understand Traceroute tool
- To understand Telnet
- To understand Netcat

Methodology

Task 1

Web Browser is discussed as a convenient tool to gather information about a target. To get access to the developer tools on firefox or google chrome, ***ctrl + shift + I*** short cut can be used. This allows us to inspect and what the browser is receiving and exchanging with the remote server.

Some add-ons were for chrome used in pen-testing were discussed which includes; **FoxyProxy, User-Agent Switcher and Manager,** and **Wappalyzer.**



```
let questions = {
    1   :   {
        'speaking'  :  'alice',
        'answer_1'  :  'SYN : Can you hear me Bob?',
        'answer_2'  :  'FIN : Goodbye',
        'answer_3'  :  'ACK : Erm... What?',
        'answer'    :   1
    },
    2   :   {
        'speaking'  :  'bob',
        'answer_1'  :  'RST : Cya Later',
        'answer_2'  :  'PING : 77',
        'answer_3'  :  'SYN/ACK : Yes, I can hear you!',
        'answer'    :   3
    },
    3   :   {
        'speaking'  :  'alice',
        'answer_1'  :  'FAIL : SEGMENTATION FAULT',
        'answer_2'  :  'ACK : Okay Great',
        'answer_3'  :  'SYN : x = 3?',
        'answer'    :   2
    },
    4   :   {
        'speaking'  :  'alice',
        'answer_1'  :  'ICMP : 99',
        'answer_2'  :  'SYN : Yes, I can hear you!',
        'answer_3'  :  'DATA : Cheesecake is on sale!',
        'answer'    :   3
    },
    5   :   {
        'speaking'  :  'bob',
        'answer_1'  :  'ACK : I Hear ya!',
        'answer_2'  :  'REPEAT : What?',
        'answer_3'  :  'RESET : Help!',
        'answer'    :   1
    },
    6   :   {
        'speaking'  :  'alice',
        'answer_1'  :  'ACK : OK',
        'answer_2'  :  'FIN/ACK : I\'m all done',
        'answer_3'  :  'ECHO : Retry',
```

```
    7   :   {
        'speaking'  :  'bob',
        'answer_1'  :  'SYN : Received',
        'answer_2'  :  'WIRE : Reset Connection',
        'answer_3'  :  'FIN/ACK : Yeah Me Too',
        'answer'    :   3
    },
    8   :   {
        'speaking'  :  'alice',
        'answer_1'  :  'SYN : Connected',
        'answer_2'  :  'ACK : Okay, Goodbye',
        'answer_3'  :  'SYN/ACK : Not Received',
        'answer'    :   2
    }
}
```

Task 2

In this level I cover ***ping,*** which is a command used to check if the remote system is online, i.e it sends a packet to the remote system and the remote system replies.

Ran '***man ping'*** command

option -s used to set the size of the data carried by ICMP echo request.

```
-S sndbuf
    Set socket sndbuf. If not specified, it is selected to buffer not more than one packet.
```

Size of ICMP header in bytes.

```
ICMP PACKET DETAILS
       An IP header without options is 20 bytes. An ICMP ECHO_REQUEST packet contains an additional 8 bytes worth of ICMP header followed by an arbitrary
       amount of data. When a packetsize is given, this indicates the size of this extra piece of data (the default is 56). Thus the amount of data
       received inside of an IP packet of type ICMP ECHO_REPLY will always be 8 bytes more than the requested data space (the ICMP header).
```

Answer the questions below

Which option would you use to set the size of the data carried by the ICMP echo request?

-s | ✓ Correct Answer | Hint

What is the size of the ICMP header in bytes?

8 | ✓ Correct Answer | Hint

Does MS Windows Firewall block ping by default? (Y/N)

Y | ✓ Correct Answer

Deploy the VM for this task and using the AttackBox terminal, issue the command `ping -c 10 10.10.117.34`. How many ping replies did you get back?

10 | ✓ Correct Answer

Task 3

Task covered ***Traceroute*** command. It is used to trace the route/hops taken by the packets from our system to another host. It's purpose was brought out as to find the IP address of the routers or hops that a packet traverses as it goes from your system to a target host. Also reveals the number of routers between systems.

Answer the questions below

In Traceroute A, what is the IP address of the last router/hop before reaching tryhackme.com?

172.67.69.208 | ✓ Correct Answer | Hint

In Traceroute B, what is the IP address of the last router/hop before reaching tryhackme.com?

104.26.11.229 | ✓ Correct Answer | Hint

In Traceroute B, how many routers are between the two systems?

26 | ✓ Correct Answer

Start the attached VM from Task 3 if it is not already started. On the AttackBox, run `traceroute 10.10.230.188`. Check how many routers/hops are there between the AttackBox and the target VM.

No answer needed | ✓ Correct Answer | Hint

Task 4

Covered Telnet (Teletype Network) which was developed in 1961 to communicate with a remote system via CLI. Telnet send all data in clear form, which makes it easier for anyone with access to the communication channel, to steal the login credentials, hence a secure alternative of SSH.

Used telnet client to connect to the VM on port 80

```
root@ip-10-10-246-131:~# telnet 10.10.154.26 80
Trying 10.10.154.26...
Connected to 10.10.154.26.
Escape character is '^]'.
get /http/1.1
HTTP/1.1 400 Bad Request
Date: Fri, 14 Jun 2024 14:03:56 GMT
Server: Apache/2.4.10 (Debian)
Content-Length: 308
Connection: close
Content-Type: text/html; charset=iso-8859-1
```

Start the attached VM from Task 3 if it is not already started. On the AttackBox, open the terminal and use the telnet client to connect to the VM on port 80. What is the name of the running server?

Apache | ✓ Correct Answer

What is the version of the running server (on port 80 of the VM)?

2.4.10 | ✓ Correct Answer

Task 5

Covered ***Netcat (nc).*** nc supports both UDP and TCP protocols. Acts as a clients that connect to a listening port and ca also act as a server that listens on a port of your choice.

A **get** is used for the default page, ***GET /HTTP/1.1***

Used nc to connect to machine IP port 21, and figured out the version of the running server.

```
root@ip-10-10-237-70:~# nc 10.10.109.122 21
220 debra2.thm.local FTP server (Version 6.4/OpenBSD/Linux-ftpd-0.17)
 ready.
```

Start the VM and open the AttackBox. Once the AttackBox loads, use Netcat to connect to the VM port 21. What is the version of the running server?

0.17

✓ Correct Answer

Conclusion

The module comprehensively covered active reconnaissance tools. The following tools were discussed; ***traceroute*** *– used to map the path to the target,* ***ping*** *– checks if the target system responds to ICMP Echo, and* ***telnet*** *– checks which ports are open and available by trying to get to them.*

In a nutshell the figure below covers all the tools discussed.

Command	Example
ping	`ping -c 10 10.10.109.122` on Linux or macOS
ping	`ping -n 10 10.10.109.122` on MS Windows
traceroute	`traceroute 10.10.109.122` on Linux or macOS
tracert	`tracert 10.10.109.122` on MS Windows
telnet	`telnet 10.10.109.122 PORT_NUMBER`
netcat as client	`nc 10.10.109.122 PORT_NUMBER`
netcat as server	`nc -lvnp PORT_NUMBER`

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Task 1 Introduction

Task 2 Web Browser

Task 3 Ping

Task 4 Traceroute

Task 5 Telnet

Task 6 Netcat

Task 7 Putting It All Together